

Erdős Problem #510

EP-510

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

If $A \subset \mathbb{Z}$ is a finite set of size N then is there some absolute constant $c > 0$ and θ such that $\sum_{n \in A} \cos(n\theta) < -cN^{1/2}$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>